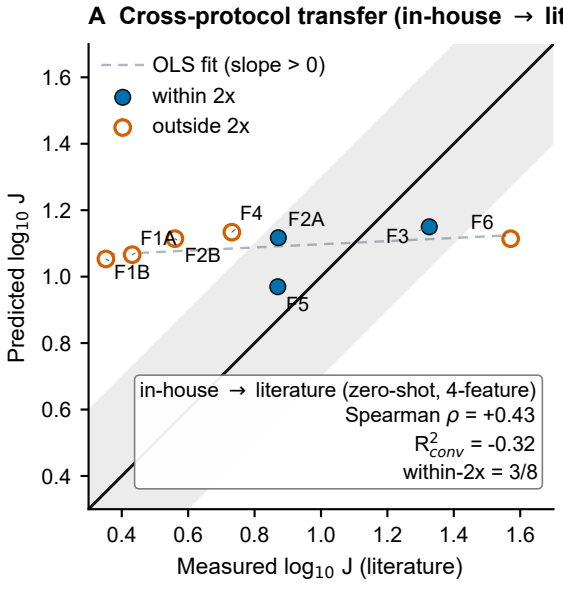
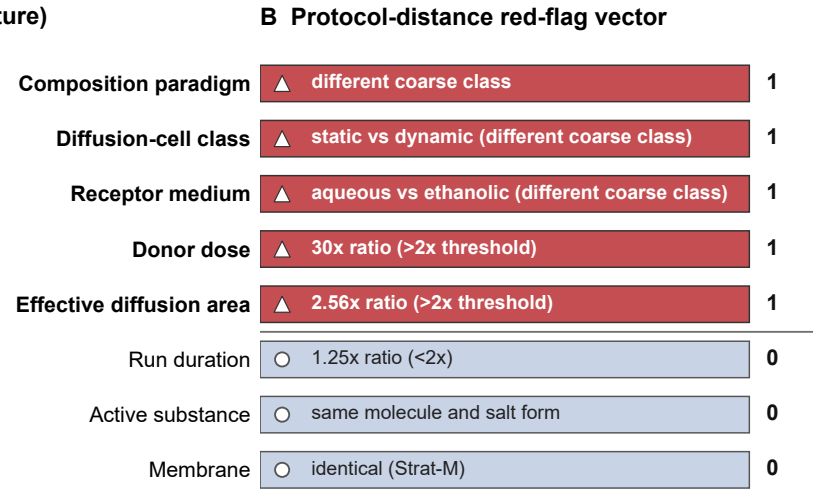

A Cross-protocol transfer (in-house → literature)
OLS fit (slope > 0)
within 2x
outside 2x
F1A
F1B
F2A
F2B
F3
F4
F5
F6
in-house → literature (zero-shot, 4-feature)
Spearman ρ = +0.43
R²conv = -0.32
within-2x = 3/8
Predicted log10 J
Measured log10 J (literature)
0.4
0.6
0.8
1.0
1.2
1.4
1.6


B Protocol-distance red-flag vector
Composition paradigm
different coarse class
1
Diffusion-cell class
static vs dynamic (different coarse class)
1
Receptor medium
aqueous vs ethanolic (different coarse class)
1
Donor dose
30x ratio (>2x threshold)
1
Effective diffusion area
2.56x ratio (>2x threshold)
1
Run duration
1.25x ratio (<2x)
0
Active substance
same molecule and salt form
0
Membrane
identical (Strat-M)
0
Continuous dimensions flagged when source/target ratio > 2x;
categorical dimensions flagged when coarse classes differ.
5 of 8 dimensions flagged.